

A Multi-Agent AI Framework for Mapping School Locations: A Case Study in Chaozhou, China

Wanyi LI, Zhongxing PENG¹, and Wei HUANG

*School of Computer Information Engineering, Hanshan Normal University,
Chaozhou, China*

Abstract. Chinas rapid urbanization has created a significant misalignment with the slower, policy-oriented allocation of educational resources, yet a critical barrier to studying this spatial inequality is the lack of high-quality, fine-grained data on different schools. This study proposes a novel multi-agent AI (Artificial Intelligence) framework to automate the construction of a standardized geospatial dataset for schools, demonstrated through a case study of Chaozhou where data from diverse sources were integrated, validated, and annotated to create a verified dataset of 638 KC12 institutions. The resulting dataset enabled the estimation of school land area, analysis of morphological characteristics, and comparison of school distribution against population density, thereby providing a robust methodology and evidence base to inform data-driven policy on educational resource allocation and advance educational equity.

Keywords. Educational Inequality, Spatial Analysis, AI Agent, School Mapping, Geographic Information Systems (GIS), Satellite Image Analysis, China Education, Resource Allocation

1. Introduction

China’s rapid urbanization has reshaped its demographic landscape, creating a persistent misalignment between the distribution of educational resources and population needs [1,2]. While a significant body of research has documented the spatial dimensions of this educational inequality, particularly in China [3,4,5,6], a critical methodological limitation remains: the reliance on heterogeneous and non-standardized geographical data on schools. Studies often depend on data from online map APIs or non-public archives [3,4,5], a practice that has been explicitly noted as a constraint on fine-grained analysis [6]. This data gap underscores the necessity for an automated, scalable method to collect reliable and precise school location data.

To address this challenge, we propose a novel multi-agent Artificial Intelligence (AI) framework. Leveraging the capabilities of large language models

¹Corresponding Author: Zhongxing Peng, zxpeng@hstc.edu.cn.

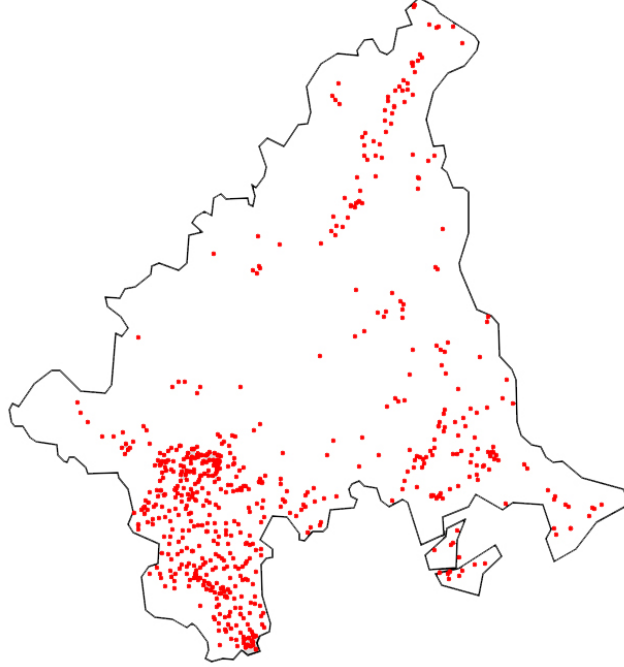


Figure 1. Schools' Map of Chaozhou City. Each red dot represents a school.

(LLMs) and AI agents [7,8,9], our framework is designed to automate the integration of information from diverse sources such as educational authorities [10] and map services to collect, verify, and standardize school-related geographical data. The system employs specialized agents to perform tasks including entity recognition, address validation, and data deduplication, culminating in the annotation of school boundaries on satellite imagery.

We demonstrate this framework by constructing a standardized dataset of 638 K-12 schools in Chaozhou. The process involves retrieving and spatially filtering point-of-interest (POI) data, followed by using the LabelMe tool to delineate polygonal boundaries for precise footprint extraction. This enables the calculation of land area and analysis of morphological features. The main contribution of this study is the proposed multi-agent AI methodology, which offers a transferable and reusable workflow for generating high-quality educational spatial data. This provides a robust evidence base for resource allocation and education policy formulation.

2. Use AI Agent to Build School Satellite Image Dataset

The data for this study were sourced exclusively from public domains, thereby precluding the need for ethical approval and ensuring the proposed pipeline and AI agents can be readily applied to other regions. The collection process began by compiling initial school lists from meta-sources like search engines and online

Table 1. Keyword Extraction Table for School Types

School Type	Keywords (Possible Terms in School Names)	Notes
Primary School	Primary School, Elementary School, Central School, Experimental Primary School, Affiliated Primary School	Clearly indicates primary-level institution
Junior High School	Junior High School, Lower Secondary School, Junior Section, Affiliated Junior High	Explicitly refers to junior high, may be part of a 9-year integrated school
Senior High School (General)	Senior High School, Middle School (only if “Junior” is not present), Advanced Middle School, Experimental Middle School, Complete Middle School	Must exclude terms like “Junior” or “Lower Secondary” to classify as senior high
Nine-Year Integrated School	Nine-Year Integrated, Nine-Year School, Experimental School, Education Group School	Includes both primary and junior high functions, requires further distinction
Complete Middle School	Complete Middle School, Advanced Middle School, Municipal Affiliated Middle School	Offers both junior and senior high education
Vocational High School	Vocational High School, Secondary Vocational School, Technical School, Vocational Education Center, Technician School, Training School	Focused on vocational education
Special Education School	Special Education, Special Needs School, School for the Deaf, School for the Blind	Provides special education services, not a regular school type
International School	International School, IB School, Foreign Nationals School, Place Names of Countries (e.g., UK, US, Canada), Bilingual School	Mostly private or schools for foreign students
Private School	Private, Independent, Experimental School (some cases), Talent/Nurturing Schools, Foreign Language School, Bilingual, Modern School	Not always level-specific, requires further judgment based on name or address
Higher Education Institution	University, College, Vocational and Technical College, Higher Vocational School	Not part of K-12, generally excluded
Kindergarten (Excluded)	Kindergarten, Nursery, Early Education Center	Belongs to preschool education, not the focus of this study, usually excluded

maps. As these raw lists often contained duplicates or were incomplete, a pre-processing stage was essential to generate a clean and comprehensive dataset for subsequent analysis.

To enhance school type identification, we developed an AI agent named STR (School Type Recognition), which utilizes raw school name and address data to perform multi-faceted validation through keyword extraction, address-augmented recognition, and database matching. STR employs predefined keywords (as shown in Table 1) to classify school types directly from names, but since name-based classification alone is often insufficient, it also incorporates geographic information from addresses — such as district or county details — to improve accuracy. For example, with a school like Chaozhou Technician College, located at No. 88, Gongye Avenue, Xiangqiao District, Chaozhou, Guangdong Province, STR extracts cues like Xiangqiao District and Gongye Avenue and aligns them with an address cue dictionary (Table-2) that maps administrative and functional labels to school

Table 2. Examples of Address-Based Keyword Cues

Category	Keywords / Cues	Notes
Industrial Area	Industrial Park, Industrial Avenue, Economic Development Zone, Technical Park, Industrial Zone	Usually where vocational high schools or technical schools are concentrated
Educational Cluster Zone	Education City, Education Park, University Town, Vocational Education Zone	Various levels of schools clustered together; further classification needed based on name
Administrative Division (County-Level)	Township, Town, Village, Remote Area, Poverty-Stricken Area	If the school name is ambiguous but the address is in a township/rural area, it is more likely to be a basic education school
Administrative Division (Urban Area)	Municipal, Urban District, City Center, Central District	Schools in the city center are often key complete secondary schools or primary schools
Special Area	Special Education Park, Rehabilitation Center, Childrens Welfare Institute	Usually refers to special education schools

Table 3. Examples of Name Deduplication by Tags of Modification

Original Name	Standardized Name	Tags of Modification	Recommendation by AI Agent
Jinshan High School Teaching Building	Jinshan High School	Remove redundant words	Merge into “Jinshan Middle School”, remove this record
Jinshan School	Jinshan High School	None	Retain
Huashi Affiliated High School	Affiliated High School of South China Normal University	Unify name	Merge into standardized name, remove abbreviation
Affiliated High School of South China Normal University	Affiliated High School of South China Normal University	Keep full name	Retain
Dongshan Primary School (South Campus)	Dongshan Primary School	Remove redundant words; manual review required	Do not process automatically, mark as review item
Chaozhou Fengxi Middle School	Chaozhou Fengxi Middle School	None	Retain

Table 4. Statistics Report of School Name Deduplication

Item	Count
Total Original Names	856
Records Retained after Deduplication	732
Records Merged	98
Processed as “Remove Redundant Words”	61
Processed as “Unify Names”	37
Processed as “Retain Full Name”	623
Processed as “Require Manual Review”	32

Table 5. School Data Structure

Field	Description
name	School name (e.g., “Jinshan High School, Chaozhou”)
location	Geographic coordinates (format: “longitude, latitude”)
address	Detailed address
tel	Contact phone number
type	school type (usually “Middle School”)
image	satellite image of a school



Figure 2. Satellite Image of Jinshan High School, Chaozhou City. **Left:** The original satellite image. **Right:** Marker Map by LabelMe.

type indicators. Further, STR integrates fuzzy matching and region-constrained strategies to cross-validate data against authoritative educational databases, such as the Guangdong Provincial Department of Education website, ensuring more comprehensive and reliable classification outcomes.

We present SNSE (School Name Standardization Expert), an AI agent designed to deduplicate and unify school names by applying a dedicated processing function that utilizes regular expression matching and string replacement to enhance consistency, clarity, and comparability. This process involves removing redundant descriptive terms (e.g., branch campus or teaching building), normalizing brackets by converting Chinese brackets to English ones, adjusting naming order (e.g., reformatting XX Affiliated XX School to XX School (Affiliated to XX)), and eliminating redundant geographical prefixes to prevent structural ambiguity and duplication. After standardization, a dual-strategy approach using containment checking and word vector similarity matching identifies potential duplicates or highly related entries, such as prefix-based overlaps or semantically similar names, which are then flagged for manual review. Each standardized record is annotated with a Tags of Modification field indicating the type of processing applied, such as removal of redundant terms, name unification, retention of the original name, or flagging for human inspection, thereby reducing data redundancy while preserving precision and authenticity.

We propose the School Name Deduplication Expert (SNDE), a specialized AI agent that performs formal data deduplication by automatically interpreting the "Tags of Modification" field to determine whether records should be retained, merged, or flagged. Records tagged for "Removal of Redundant Terms" or "Name Unification" are automatically merged to preserve a single standardized entry, while those marked for "Retention of Original Name" are kept intact. To ensure precision, records requiring "Manual Review" are flagged for human intervention, creating a semi-automated workflow that balances efficiency with oversight. Furthermore, when standardization results in identical names, the agent proactively suggests merging them, thereby establishing a high-quality, standardized school

list through a coordinated blend of automated processing and manual verification, as illustrated in Table 3.

Statistics report of school name deduplication is showed in Table-4.

Leveraging the school list, we employed the Gaode API [11] (<https://amap.com/>) to retrieve open-access satellite imagery for each school, with the data structure formally defined in Table-5; the entire workflow was automated through Python scripts that handled API requests, response parsing, and data storage, facilitating efficient and standardized collection of primary school geospatial data in Chaozhou, and using the acquired geographic coordinates, we generated a spatial distribution map of schools in Chaozhou city, as illustrated in Figure-1.

Given the potential inaccuracies in open geographic data, we enhanced the precision of spatial coordinates by integrating GIS data from multiple online mapping platforms, including Tianditu [12] (<https://www.tianditu.gov.cn/>) and Gaode. To ensure accurate alignment, four key coordinate points along the boundary of Xunzhou Island were selected, and satellite imagery of the area was acquired from both sources; using the LabelMe tool [13] (<https://github.com/wkentaro/labelme>), three representative key points were annotated on the corresponding images from each platform, and the latitude and longitude offsets between them were calculated programmatically in Python to derive an overall correction value. Finally, based on the computed offsets, the original school distribution map of Chaozhou was adjusted by correcting the latitude and longitude of each boundary dataset, resulting in a geographically accurate map suitable for further visualization and spatial analysis.

Based on the constructed dataset, we manually annotated the boundaries of all 638 K-12 schools in Chaozhou City using the open-source tool LabelMe, which was selected for its precise polygonal annotation capabilities and web-based collaborative interface, enabling efficient large-scale labeling. Annotators meticulously drew polygons along school fences or natural boundaries visible on high-resolution satellite images, with the resulting dataset comprising two main parts: the raw satellite imagery and corresponding JSON files in the LabelMe standard format that record the polygonal vertex coordinates for each school, as exemplified in Figure-2.

3. Applications of School Satellite Image Dataset

This section presents several applications of the proposed dataset, including estimating school land area, analyzing school morphological features, and comparing the spatial distributions of population and schools.

3.1. Estimate School Land Area

First, the polygon area in pixels is calculated using the Shoelace Formula [14]. Suppose a polygon in a LabelMe annotation file has n vertices with pixel coordinates $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. The Shoelace Formula is applied to compute the pixel area A_{px} of the polygon as follows:

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$$A_{\text{px}} = \frac{1}{2} |(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + \cdots + (x_ny_1 - x_1y_n)|. \quad (1)$$

The shoelace formula computes the area of a closed polygon by summing the signed areas of parallelograms formed by consecutive edges, with the final area obtained by taking the absolute value of this sum and dividing by two. The procedure involves listing the vertex coordinates sequentially while appending the first vertex at the end, then for each consecutive vertex pair calculating $x_iy_{i+1} - x_{i+1}y_i$, summing all these differences, taking the absolute value of the sum, and dividing by two.

Second, pixel measurements are converted to actual ground distances. In web maps, the meters-per-pixel (mpp) ratio depends on the zoom level $zoom$ and latitude φ . The conversion is given by:

$$\text{mpp} = \cos(\varphi) \cdot \frac{2\pi R}{256 \cdot 2^{\text{zoom}}}, \quad (2)$$

where $R = 6378137$ m is Earth's mean radius, and $256 \cdot 2^{\text{zoom}}$ is the pixel width of the world map at the given zoom level. The $\cos(\varphi)$ term corrects for east-west scaling at different latitudes. For example, at central Chaozhou ($\varphi \approx 23.67^\circ$) and $zoom = 18$, we obtain $\text{mpp} \approx 0.597$ m/px.

Third, the actual area in square meters is calculated from the pixel area A_{px} using:

$$A_{\text{m}^2} = A_{\text{px}} \times (\text{mpp})^2. \quad (3)$$

Fourth, if a school is represented by multiple non-contiguous polygons, the total school area is the sum of the areas of all m constituent polygons:

$$A_{\text{school}} = \sum_{k=1}^m A_{\text{m}^2}^{(k)}. \quad (4)$$

The estimated land areas of schools in Chaozhou vary considerably by type, with primary schools generally occupying 2,000 to 15,000 m² and secondary schools ranging from 10,000 to 40,000 m², reflecting differing infrastructure needs across educational stages. Most primary schools fall in the medium range of 3,000-10,000 m², while secondary schools are typically larger, commonly spanning 15,000-40,000 m². Overall, school land areas follow a clear hierarchical pattern: high schools occupy the largest areas (Figure-4), followed by complete schools, junior high schools, nine-year integrated schools, and primary schools (Figure-3), reflecting increasing spatial demands at higher educational levels. Urban-rural disparities are also evident, though some central township schools rival urban ones in scale. Experimental or newly built private schools tend to have larger areas, indicating a trend toward scale and modernization, while branch campuses usually have smaller footprints, reflecting their distinct functional positioning.

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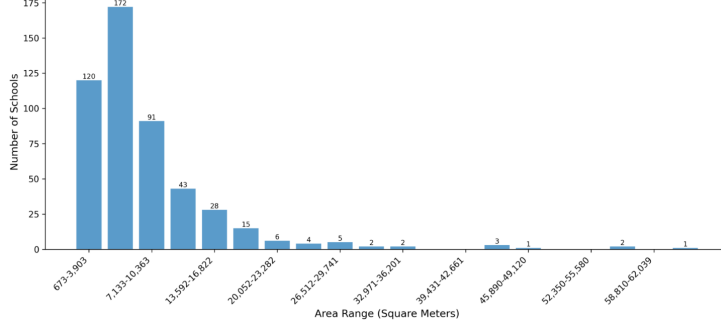


Figure 3. Numbers of Primary Schools with Different Areas

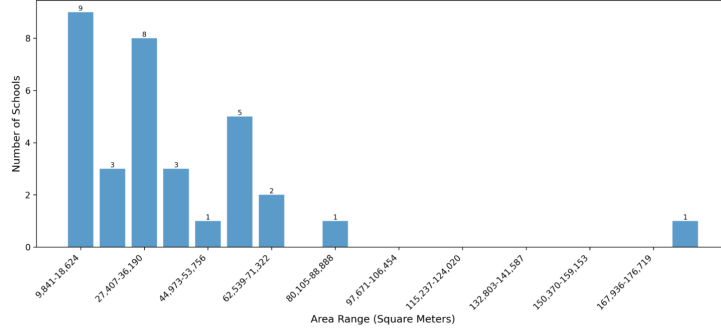


Figure 4. Numbers of High Schools with Different Areas

3.2. Analyze School Morphological Features

The morphological features of the schools are characterized by a 7-dimensional vector derived from the Hu invariant moments computed for each school contour, representing a school as $\mathbf{x}_i \in \mathbb{R}^7$. To explore potential distribution patterns of school contours based on shape features, we applied K-means clustering, which minimizes the sum of squared distances between points and their cluster centroids with following objective:

$$\min_C \sum_{k=1}^K \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2, \quad (5)$$

where K is the number of clusters, C_k denotes the k -th cluster, and $\boldsymbol{\mu}_k$ is the centroid of cluster k . In our experiment, we set $K = 8$.

The clustering results were visualized using t-distributed Stochastic Neighbor Embedding (t-SNE), which minimizes the Kullback-Leibler divergence between two similarity distributions:

$$KL(P||Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}, \quad (6)$$

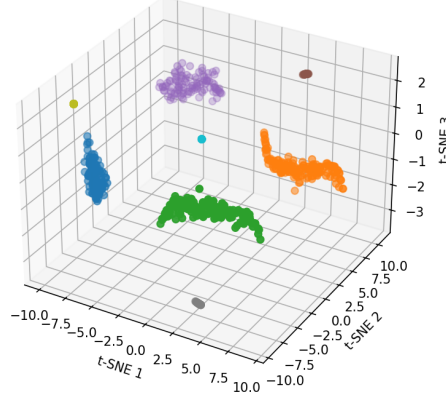


Figure 5. 3D t-SNE Projection of KMeans ($K = 8$) Clustering Results (perplexity = 75)

where p_{ij} and q_{ij} represent the pairwise similarities in the high-dimensional and low-dimensional spaces, respectively. By adjusting the perplexity parameter, t-SNE balances the preservation of local and global data structures. Figure-5 illustrates the three-dimensional embeddings obtained with a perplexity of 75, where distinct colors correspond to different K-means clusters. The visualization reveals that while some clusters are well-separated indicating discriminative Hu moment features others exhibit blurred boundaries, suggesting considerable morphological similarities among schools from different clusters. Overall, the combined use of K-means and t-SNE provides an intuitive visualization of school morphology, effectively highlighting both distinctions and overlaps in contour shapes.

3.3. Compare the Spatial Distributions of Population and Schools

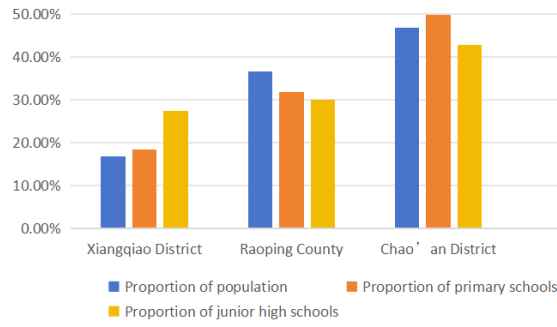


Figure 6. Comparison of the Proportion of Births and the Number of Primary and Secondary Schools in the Three Major Districts of Chaozhou (Chao'an, Raoping, and Xiangqiao) in 2023. Data source: *Chaozhou Statistical Yearbook 2024*.

Based on data from the Chaozhou Statistical Yearbook 2024 [15], a comparison between birth proportions and the distribution of primary and secondary schools across the city's main districts reveals notable patterns: Chao'an District

accounts for 46.78% of births and holds 49.7% of primary schools and 42.7% of secondary schools, indicating a strong alignment between its demographic weight and educational resources; Raoping, with 36.48% of births, has 31.8% of primary and 30.0% of secondary schools, suggesting potential resource pressure relative to its population; while Xiangqiao, with 16.73% of births, hosts 18.5% of primary and a higher 27.3% of secondary schools, reflecting a greater concentration of secondary educational infrastructure in the urban core. These disparities are visualized in bar charts (Figure 6), offering a data-driven basis for educational planning and demographic analysis.

4. Conclusion

This study makes a significant theoretical contribution by introducing a novel multi-agent AI framework that bridges a critical methodological gap in educational equity research: the transition from point-based representations of schools to quantifiable areal units derived from satellite imagery. This framework, which automates the construction of a precise, high-resolution geographical dataset, provides a transferable and scalable solution to the field's longstanding reliance on non-standardized data, thereby enhancing the reliability and analytical granularity of spatial analysis. The empirical application in Chaozhou demonstrates the framework's utility in uncovering nuanced patterns of inequality through metrics like land area and spatial mismatch, offering a powerful tool for evidence-based policy. While the study's focus on a single city establishes a robust proof of concept, it also presents a limitation regarding the generalizability of the findings, highlighting the need for future validation across diverse urban contexts. Subsequent research can build on this foundation by expanding the framework's scope to include multi-city comparisons, temporal analysis, and deeper AI integration for dynamic monitoring of educational resources.

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